

What is DAX?

DAX (Data Analysis Expressions) is a formula language used in Power BI, Excel, and other Microsoft tools for creating custom calculations on your data. Think of it as the "Excel formulas" for Power BI, but more powerful and designed for data modeling.

Why is DAX Important?

In Power BI, the data you load often needs to be **transformed or calculated** to generate meaningful insights. DAX lets you:

- Create calculated columns and measures
- Build KPIs and advanced metrics
- Work with time-based data (like YTD, MTD)
- Apply filters and context to data analysis

DAX Components

There are three main places where you use DAX:

Component	Description
Calculated Column	Adds a new column to a table, evaluated row by row. Stored in the model.
Measure	Calculates a value based on filters or user interaction. Calculated on the fly.
Calculated Table	Creates a new table from existing tables/data using DAX.

Basic DAX Syntax

- DAX functions look like Excel formulas:

Total Sales = SUM(Sales[SalesAmount])

- DAX is **case-insensitive** but it's good practice to use proper formatting.

Commonly Used DAX Functions

Category	Function	Example	Description
Aggregation	SUM, AVERAGE, MIN, MAX, COUNT	SUM(Sales[Amount])	Basic math
Logical	IF, AND, OR, SWITCH	IF(Sales[Amount] > 1000, "High", "Low")	Conditions
Text	CONCATENATE, LEFT, RIGHT, SEARCH	CONCATENATE(Customer[FirstName], " ", Customer[LastName])	String operations
Date/Time	TODAY, YEAR, MONTH, DATESYTD, DATEADD	YEAR(Orders[OrderDate])	Date analysis
Filter	CALCULATE, FILTER, ALL	CALCULATE(SUM(Sales[Amount]), Region="West")	Context-based calc

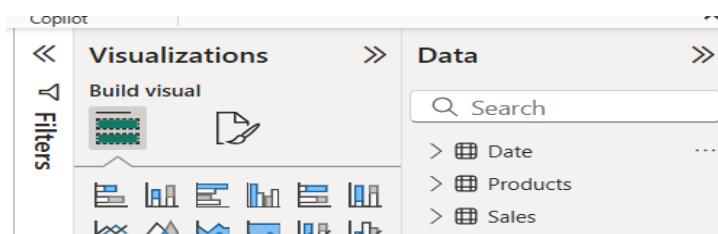
Category	Function	Example	Description
Time Intelligence	TOTALYTD, SAMEPERIODLASTYEAR, PARALLELPERIOD	TOTALYTD(SUM(Sales[Amount]), 'Date'[Date])	Period-over-period

Calculated Column vs Measure: Key Difference

Feature	Calculated Column	Measure
Row Context	Evaluated for every row	Calculated for aggregation level
Storage	Stored in memory	Not stored
Use Case	Categorization, derived fields	KPI, metrics, total values
Performance	Slower on large data	Faster

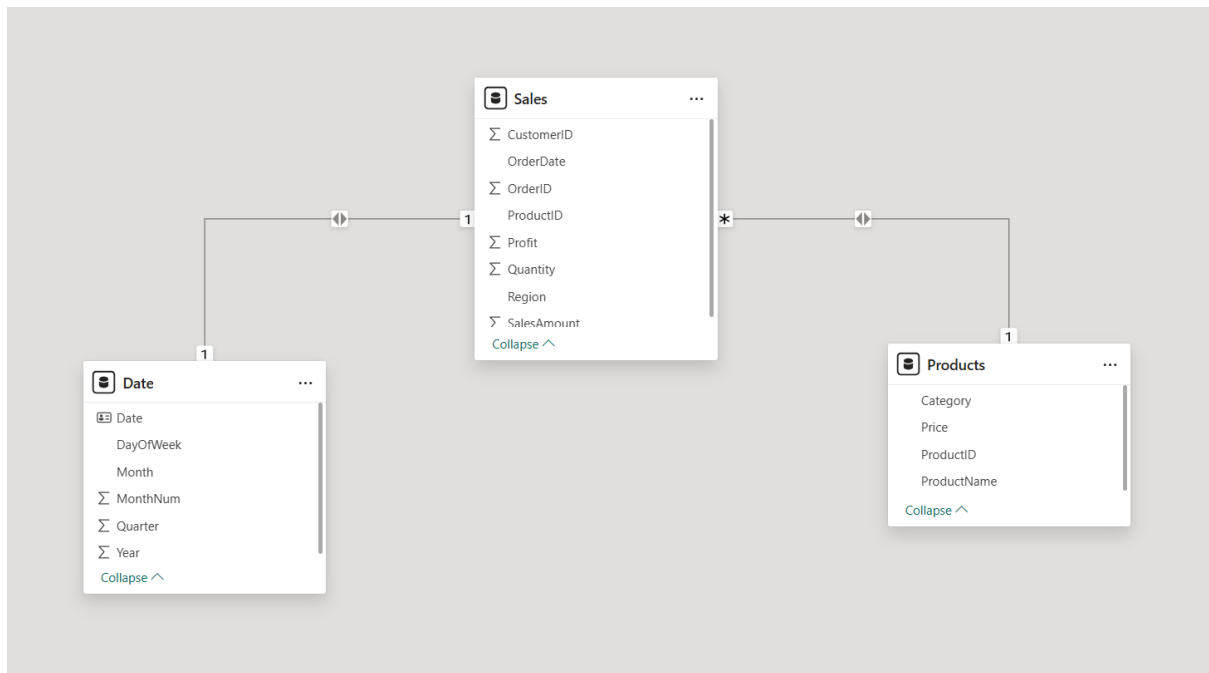
STEP 1: Load Data into Power BI

1. **Open Power BI Desktop.**
2. Click **Home > Get Data > Excel.**
3. Select the file:
powerbi_dax_sample_data.xlsx
4. Import all three sheets:
 - Sales
 - Products
 - Date
5. Click **Load.**



STEP 2: Create Relationships

1. Go to **Model View** (icon on left panel).
2. Create these relationships:
 - Sales[ProductID] → Products[ProductID]
 - Sales[OrderDate] → Date[Date]
3. Right-click the **Date table** → Mark as Date Table → select Date.

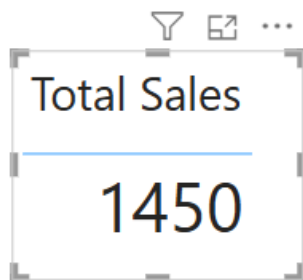


STEP 3: Create DAX Measures

Go to **Modeling > New Measure** and add these:

1. Total Sales

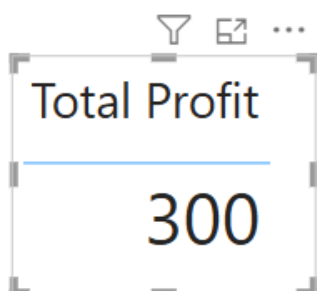
```
1 Total Sales = SUM(Sales[SalesAmount])
```



This screenshot shows the Power BI interface for the 'Total Sales' measure. In the Columns pane, 'Total Sales' is selected. In the Fields pane, 'SalesAmount' is unchecked, and 'Total Sales' is checked. The 'Drill through' section shows 'Cross-report' set to 'Off' and 'Keep all filters' set to 'On'.

2. Total Profit

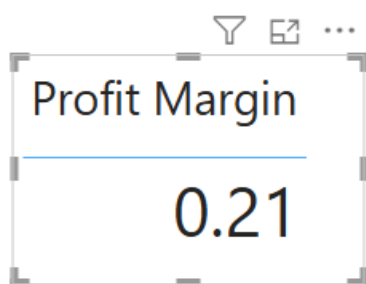
```
1 Total Profit = SUM(Sales[Profit])
```



This screenshot shows the Power BI interface for the 'Total Profit' measure. In the Columns pane, 'Total Profit' is selected. In the Fields pane, 'SalesAmount' is unchecked, 'Total Profit' is checked, and 'Total Sales' is unchecked. The 'Drill through' section shows 'Cross-report' set to 'Off'.

3. Profit Margin

```
1 Profit Margin = DIVIDE([Total Profit], [Total Sales])
```



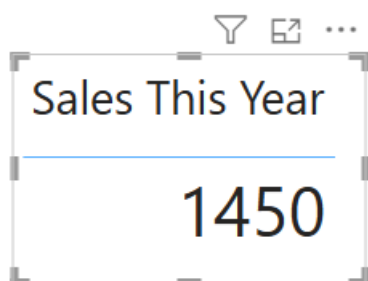
Visual Studio Code interface showing the 'Columns' pane. The 'Profit Margin' field is selected in the 'Columns' list. The 'Drill through' section is visible below the 'Columns' list.

Fields in the 'Columns' list:

- ☐ ProductID
- ☐ Σ Profit
- ☒ Profit Margin
- ☐ Σ Quantity
- ☐ Region
- ☐ Σ SalesAmount
- ☐ Total Profit
- ☐ Total Sales

4. Sales This Year

```
1 Sales This Year = CALCULATE([Total Sales], DATESYTD('Date'[Date]))
```



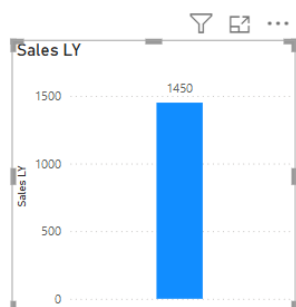
Visual Studio Code interface showing the 'Columns' pane. The 'Sales This Year' field is selected in the 'Columns' list. The 'Cross-report' toggle is set to 'Off'.

Fields in the 'Columns' list:

- ☐ ProductID
- ☐ Σ Profit
- ☐ Profit Margin
- ☐ Σ Quantity
- ☐ Region
- ☒ Sales This Year
- ☐ Σ SalesAmount
- ☐ Total Profit
- ☐ Total Sales

5. Sales Last Year

```
1 Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
```



Visual Studio Code interface showing the 'Columns' pane. The 'Sales LY' field is selected in the 'Columns' list. The 'X-axis' and 'Y-axis' sections are visible below the 'Columns' list.

Fields in the 'Columns' list:

- ☐ Σ CustomerID
- ☐ OrderDate
- ☐ Σ OrderID
- ☐ ProductID
- ☐ Σ Profit
- ☐ Σ Quantity
- ☐ Region
- ☒ Sales LY
- ☐ Σ SalesAmount
- ☐ Test LY
- ☐ Total Sales

6. YOY Growth

```
1 YOY Growth = DIVIDE([Total Sales] - [Sales LY], [Sales LY])
```

YOY Growth

0.00

Filters on this page

Add data fields here

Filters on all pages

Add data fields here

Grid

Values

Column headers

Text

Font

☐ Σ Profit

☐ Σ Quantity

☐ Region

☐ Sales LY

☐ SalesAmount

☐ Total Sales

☒ YOY Growth

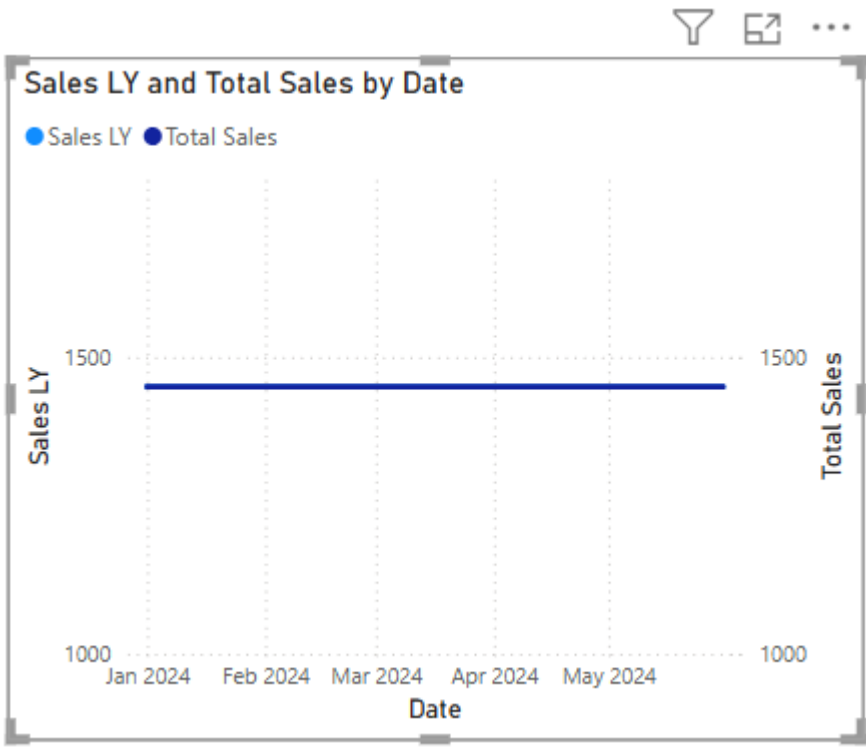
STEP 4: Build the Dashboard

Page 1 – Sales Overview

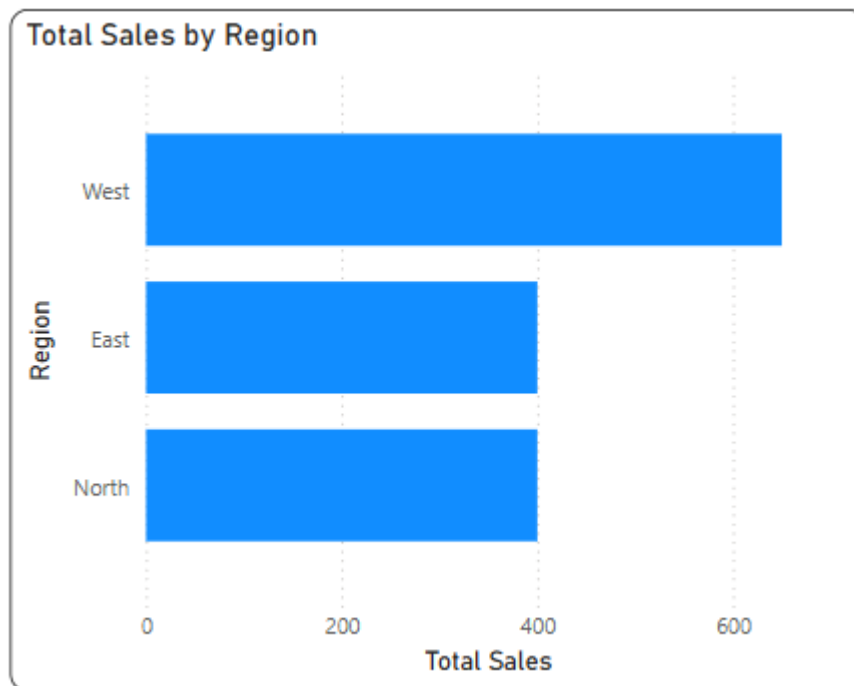
- KPI Cards for:



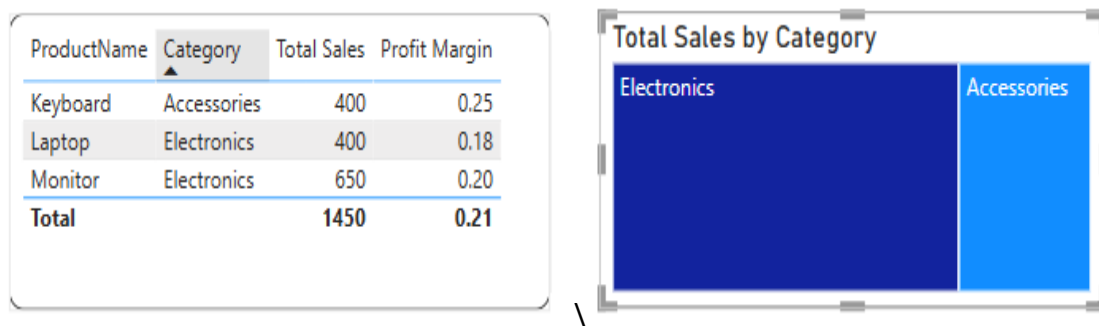
- Line Chart:



- Bar Chart:



Page 2 – Product Performance



Page 3 – Time Intelligence

